Ligjerata 4d

Impulsi i forcës

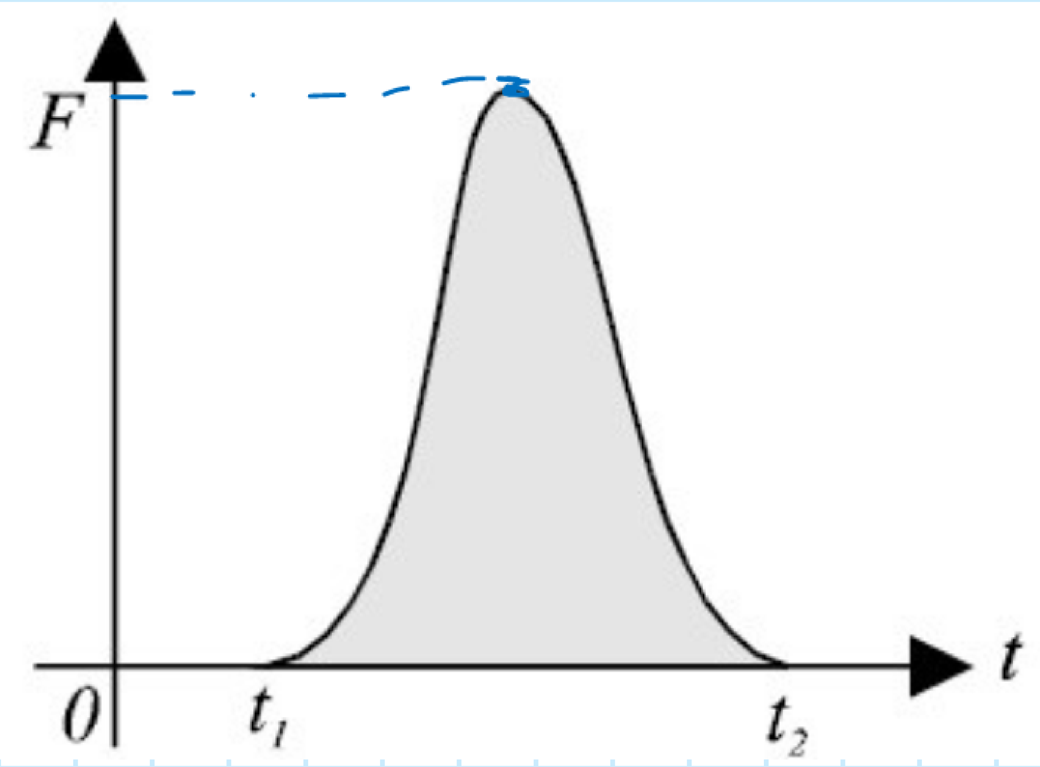
$$F = ma = m\frac{dv}{dt} \quad / \cdot dt$$

$$F \cdot dt = m\,dv \quad / \int$$

$$\int_{t_1}^{t_2} F\,dt = \int_{v_0}^{v} m\,dv$$

$$\int_{t_1}^{t_2} F\,dt = m\left(v\Big|_{v_0}^{v}\right)$$

$$\int_{t_1}^{t_2} F\,dt = m(v - v_0)$$



$$\int_{t_1}^{t_2} F\,dt = F(t_2 - t_1) = i$$

$$F\,\Delta t = i = N \cdot s = kg\,\frac{m}{s^{\cancel{2}}}\cdot \cancel{s} = \underline{kg\,\frac{m}{s}}$$

Forca e rëndimit

$$F = m\frac{dv}{dt} = \text{konst}$$

$$F \cdot dt = m\,\cancel{dv} \quad / : m$$

$$\frac{F}{m} \cdot dt = dv \quad / \int$$

$$\frac{F}{m}\int_0^t dt = \int_{v_0}^{v} dv$$

$$\boxed{\frac{F}{m} = a = g}$$

$$g\left(t\Big|_0^t\right) = v\Big|_{v_0}^{v}$$

$$gt = v - v_0$$

$$\boxed{v = v_0 + gt} \qquad \boxed{F = mg}$$